# MINI-PROJECT SHORT TECHNICAL REPORT

**Course:** Cross-Platform Mobile App Development (VKU)

**Mini-Project Title:** Mini-Project 3: OCR Expense Tracker & Receipt Parser (Flutter 3.x & Dart 3)

**Team / Student Name:** Nguyễn Thị Thương

**Submission Date:** 12/10/2026

## 1. GENERAL INFORMATION & DELIVERABLE LINKS

* **Team Members:**

1. **Nguyễn Thị Thương** — Student ID: **23IT.B219** — Class: **23SE4** — Course Class: **Phát triển ứng dụng di động đa nền tảng (4)** — Instructor: **TS. Nguyễn Thanh Tuấn** — Role: **Full-Stack Mobile & On-Device AI Developer** — Contribution: **100%**

* 🔗 **Live Demo URL:** https://vku-ocr-expense.thuong2213.workers.dev/

* 📱 **Signed Release APK (`app-release.apk`):** https://github.com/thuong614oh65/vku-ocr-expense-tracker/releases/download/v1.0.0/app-release.apk

* 💻 **GitHub Repository:** https://github.com/thuong614oh65/vku-ocr-expense-tracker

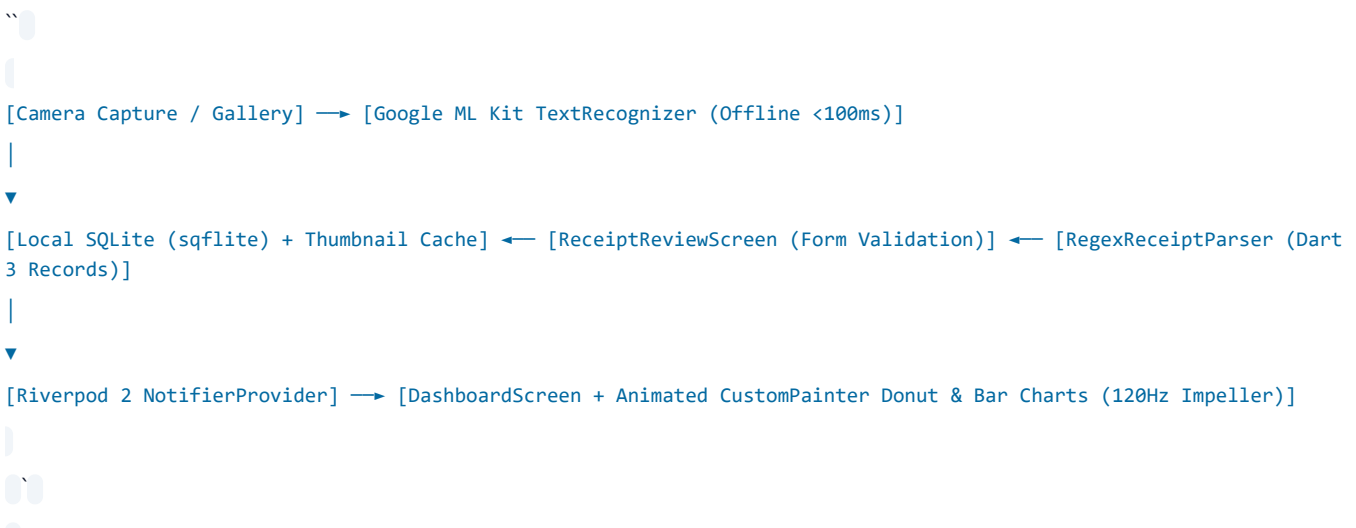
## 2. FEATURE IMPLEMENTATION CHECKLIST (10-POINT RUBRIC ALIGNMENT)

| # | Required Feature | Status | Implementation Details & Acceptance Level |
|---|---|---|---|
| 1 | **Camera Capture, Focus Tap & Framing Crop** | ✅ Complete | `CameraScannerScreen` + `CameraOverlayPainter`: Live camera viewfinder with Flash/Torch toggle, tap-to-focus (`AF`) reticle animation, and even-odd framing crop overlay with animated laser. |
| 2 | **On-Device Text Recognition (ML Kit)** | ✅ Complete | `OcrService` integrates `google_mlkit_text_recognition` (`TextRecognizer`) for offline, sub-100ms on-device text extraction with zero cloud cost. |
| 3 | **Regex Heuristic Parser & Dart 3 Records** | ✅ Complete | `RegexReceiptParser` uses multi-stage regex heuristics & Dart 3 Records `(String merchant, double total, DateTime date)` to extract Vietnamese totals (`150.000 đ`, `150,000 VND`), dates (`DD/MM/YYYY`), merchants, and 6 categories. |
| 4 | **Interactive Review & Form Validation** | ✅ Complete | `ReceiptReviewScreen` uses `Form`, `GlobalKey`, `AutovalidateMode.onUserInteraction`, `TextFormField` validators, `FocusNode` chaining, AI confidence badge, and raw OCR inspector. |
| 5 | **Riverpod 2 Compile-Safe State Management** | ✅ Complete | `expense_provider.dart` implements `ExpenseController extends Notifier<>`, `ThemeModeController`, and `CategoryFilterController` consumed via `ConsumerStatefulWidget`. |
| 6 | **Local SQLite (`sqflite`) & Thumbnail Cache** | ✅ Complete | `StorageService` persists transactions in `sqflite` (`vku_expense_ocr.db`) with Web `SharedPreferences` fallback, and caches cropped receipt thumbnails in the app documents directory (`path_provider`). |
| 7 | **CustomPainter Animated Donut / Pie Chart** | ✅ Complete | `CustomPieChart` draws category distribution directly on GPU Canvas via `canvas.drawArc` + `AnimationController` (`Curves.easeOutCubic`) without any third-party chart library. |
| 8 | **CustomPainter Animated Weekly Bar Chart** | ✅ Complete | `CustomBarChart` renders 7-day (`T2 → CN`) spending bars using `canvas.drawRRect`, `LinearGradient`, dashed grid lines, and `TextPainter` compact currency labels. |
| 9 | **Material 3 Light / Dark Mode & Polish** | ✅ Complete | Full Material Design 3 seed color harmonization (`#0284C7`), one-tap Light/Dark mode toggle, and `ExpenseSummaryCard` wrapped in `Dismissible(key: ValueKey(txn.id))`. |

| # | Required Feature | Status | Implementation Details & Acceptance Level |
|---|---|---|---|
| 10 | **Declarative Routing & Platform Channel** | ✅ Complete | `GoRouter` with `ShellRoute` persistent bottom navigation ( `app_router.dart` ) and `MethodChannel('vn.edu.vku/device_info')` ( `device_info_channel.dart` ) for native battery/haptic interop. |

## 3. TECHNICAL ARCHITECTURE & OCR REGEX TUNING

### 3.1. End-to-End System Pipeline & Clean Architecture




* **Modular Directory Structure:** lib/core/ ( app_theme.dart , app_router.dart ) • lib/models/ ( expense_category.dart , receipt_transaction.dart ) • lib/services/ ( ocr_service.dart , regex_parser_service.dart , storage_service.dart , device_info_channel.dart , mock_receipts_service.dart ) • lib/state/ ( expense_provider.dart ) • lib/widgets/ ( custom_pie_chart.dart , custom_bar_chart.dart , camera_overlay_painter.dart , expense_summary_card.dart ) • lib/screens/ ( dashboard_screen.dart , camera_scanner_screen.dart , receipt_review_screen.dart ).

### 3.2. Real-World Vietnamese Receipt Regex Tuning Table

| Target Field | Regular Expression Pattern | Edge Cases Handled |
|---|---|---|
| **Total Keyword Line** | `(?:tổng\s*(?:cộng\|tiền\|thanh\stoán)?\|thanh\stoán\|phải\strả\|total\|amount\sdue)[\s:=-]*([0-9]{1,3}(?:[.,][0-9]{3})*\|[0-9]+)` | Matches TỔNG CỘNG: 87,500 VND , THANH TOÁN: 115.000 đ , TOTAL: 245000 across noisy thermal prints. |
| **Currency Number Normalization** | `\b([0-9]{1,3}(?:[.,][0-9]{3})+)\s*(?:đ\|vnd\|vnđ)?\b` | Normalizes both dot-separated (150.000 ) and comma-separated ( 150,000 ) Vietnamese thousands, while rejecting 10-digit phone numbers and Tax IDs ( MST ). |
| **Transaction Date (DD/MM/YYYY )** | `\b(0-3]?[0-9])[/\-./\-.\b` | Extracts 28/09/2026 , 12-10-2026 , 12.10.26 and pairs with ([0-2]?[0-9]):([0-5][0-9]) timestamp. |
| **Merchant Header & Category** | Dictionary lookup (WinMart+ , Highlands Coffee , Fahasa , Petrolimex , Thế Giới Di Động ) + Top-4 header line heuristic | Filters out generic header words (HÓA ĐƠN , PHIẾU THANH TOÁN , ĐỊA CHỈ , MST ) and auto-maps to 6 categories via Dart 3 switch expressions. |

## 4. EMPIRICAL EVIDENCE & SCREENSHOTS (LIGHT & DARK MODES)

* **Evidence 1 — Dashboard in Light & Dark Mode (DashboardScreen):** Displays VKU student header (Nguyễn Thị Thương • 23IT.B219 • 23SE4), Light/Dark theme toggle (🌙 / 🌞), KPI cards (Total Spent & 3,500,000 ₫ Monthly Budget), Category Filter chips, and virtualized ExpenseSummaryCard list with swipe-to-delete (Dismissible).

* **Evidence 2 — Custom Canvas Visualizations (CustomPainter):** Hardware-accelerated Donut Pie Chart (canvas.drawArc) with animated sweep angle and center total counter, paired with 7-day Weekly Bar Chart (canvas.drawRRect) highlighting peak spending days in VKU Sky Blue gradient.

* **Evidence 3 — Live Camera Viewfinder & Framing Overlay (CameraScannerScreen):** Real-time camera feed with 4 glowing corner brackets, animated scanning laser, tap-to-focus (AF) reticle, Flash toggle, and 5 one-tap VKU sample receipt chips (WinMart+, Highlands, Fahasa, Petrolimex, Thế Giới Di Động).

* **Evidence 4 — Interactive OCR Review & Manual Correction (ReceiptReviewScreen):** Shows AI Confidence badge (96% • 24ms), cached receipt thumbnail preview, validated TextFormField inputs for Merchant, Total Amount (VND), Date, Category ChoiceChip selector, and expandable Raw OCR text viewer.

## 5. TECHNICAL CHALLENGES, RESOLUTIONS & LIMITATIONS

1. **Challenge 1 — Disambiguating Receipt Totals from Phone Numbers & Tax IDs (MST):** Vietnamese receipts contain 10-digit phone numbers (02363667888), tax codes (0104918404), and cash tendered (TIỀN MẶT: 100,000). *Resolution:* Prioritized context-anchored regex lines (TỔNG CỘNG, PHẢI TRẢ) before falling back to max-number heuristics, and filtered out unformatted $\ge 10$-digit integers.

2. **Challenge 2 — Zero-Jank 120Hz Custom Chart Animations Without Third-Party Packages:** *Resolution:* Implemented custom shouldRepaint() checks in _DonutChartPainter and _WeeklyBarChartPainter driven by CurvedAnimation(curve: Curves.easeOutCubic), issuing direct drawArc and drawRRect` calls to Flutter's Skia/Impeller canvas.

3. **Current Limitations & Future Work:** Extremely crumpled or faded thermal receipts under low light can degrade OCR character accuracy on Vietnamese diacritics; future iterations will add OpenCV adaptive thresholding pre-processing and multi-page receipt stitching.

*Đà Nẵng, Ngày 12 tháng 10 năm 2026*

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